MIL-P-28659A(YD)
31 October 1979
SUPERSEDING
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28 February 1973

MILITARY SPECIFICATION

PUBLIC ADDRESS SET, SOLID STATE, 100 WATT

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes a public address set consisting of one solid-state amplifier, capable of producing 100 watts or less of output signal power, four trumpet speakers complete with drivers, two paging speakers, one dynamic microphone with floor stand, and accessory terminal fittings and hardware. Power sources shall be 105 to 125 volts alternating current (ac), single phase, and 60 Hertz (Hz).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116 - Preservation-Packaging, Methods of.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic and Associated Equipment).
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5830

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STANDARD

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

WD-1 - General-Purpose Wiring Devices.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

UNDERWRITERS' LABORATORIES, INC., (UL)

UL813 -Sound-Recording and Reproducing Equipment, Electric.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

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3. REQUIREMENTS

* 3.1 Design. The design, when specified (see 6.2.1), of the public address set shall be in accordance with good engineering practice and shall provide accessibility for maintenance and service in the field. The design shall preclude conditions which may be hazardous to personnel or deleterious to equipment and shall be in accordance with UL813.

3.2 Standard commercial product. The public address set shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the public address sets being furnished. Standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs or brochures, and represents the latest production model(s).

* 3.3 Identical items. Public address sets of the same classification furnished under any specific contract shall be physically and mechanically identical. This requirement includes parts, assemblies, components, and accessories. No deviation will be acceptable without prior written approval of the contracting officer.

* 3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specifically specified.

* 3.5 Construction. The public address set shall be so designed and constructed to allow for easy connection and installation. All receptacles, jacks, terminals, and controls shall be labeled for proper identification of each respectively. The housing of the amplifier shall be properly grounded to prevent a shock hazard to personnel. The public address set shall be rigidly constructed to withstand adverse conditions.

3.6 Amplifier. The amplifier shall be an all solid-state design capable of supplying output power at 25 volt or 70.7 volt speaker line at various impedance to match speaker impedance. Power supply for the amplifier shall be integral with the amplifier chassis and shall be isolated from the power supply input by a transformer. The amplifier shall be wired so that the microphone has precedence over other inputs. The microphone jack shall be of the type "XLR" to be compatible with the microphone connector.

3.6.1 Signal inputs. The amplifier shall be capable of accepting signals from a low impedance microphone, a high impedance microphone, and a high impedance auxiliary input.

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* 3.6.2 Outputs. The amplifier shall have at least four output terminals. The amplifier shall have safety circuits to protect the amplifier from damage when the outputs are shorted or open.

* 3.6.3 Response. Frequency response shall be flat to within ±2.0 decibils (dB) over a range of 80 to 12,000 Hz, at full rated output power into rated load impedance.

3.6.4 Distortion. Total harmonic distortion shall be less than 3 percent at rated power and over the frequency response range.

* 3.6.5 Hum and noise. Hum and noise (at rated output) shall be no greater than -65 dB for microphone and -70 dB for auxiliary.

* 3.6.6 Controls. Amplifier controls shall include as follows: A power ON-OFF switch with an associated pilot light indicator; independent input volume level controls for each input; a master volume control; a treble and bass control.

* 3.6.7 Power cord. The amplifier shall be designed to operate at 120 volt, ac, 60 Hz. The cord shall be two conductor with ground. The power source end shall have a parallel blade grounded type connector rated for 15 amperes, 120 volts, ac, in accordance with NEMA WD-1, configuration 5-15P. The cord shall be type SJO or SJTO, with No. 18 American Wire Gage or larger stranded copper conductors. Connector and cord shall bear the UL listing mark.

3.7 Loudspeakers. Construction shall be weatherproof for outdoor application. Each loudspeaker shall be equipped with an adjustable mounting bracket.

* 3.7.1 Horns. Type and styles of horns shall be of the standard product manufactured by the company. Each horn shall have an overall dispersion angle of at least 85°. Low frequency cutoff shall not be greater than 300 Hz. The horn shall have a built-in driver or means to mount a driver unit compatible with the horn.

* 3.7.2 Horn drivers. Each driver shall incorporate an impedance matching capability for compatibility with the public address set amplifier output impedance. Driver power rating shall be adequate for operation without damage to the speakers at maximum driver output power. The frequency response shall be flat within ±3 dB from 300 Hz to 12000 Hz.

* 3.7.3 Paging loudspeakers. Each paging loudspeaker shall exhibit a dispersion angle not less than 110°, and shall have a frequency response which is flat within ±3 dB from 300 Hz to 12000 Hz. The loudspeaker impedance shall be matched with amplifier output impedance for maximum output power and efficiency.

* 3.8 Microphone. The microphone shall be a dynamic type compatible with the public address set amplifier and shall be complete with an ON-OFF switch, not less than 24 feet of shielded cable with a type "XLR" male connector, and an adjustable height floor stand which provides for variable tilting of the microphone. The pickup pattern shall be semidirectional or cardioid. The frequency response shall be flat within ±3 dB over the range of 80-12000 Hz.

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3.9 Wiring diagram. A wiring diagram on the public address set and of the amplifier shall be suitably protected and attached within or on to the amplifier case, and shall be easily accessible for maintenance reference purposes.

* 3.10 Fungus resistance. When specified (see 6.2.1), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

(a) Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
(b) Current-carrying contact surfaces, such as relay contact points, shall not be coated.

3.11 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. Unless otherwise specified (see 6.2.1), the color of the finish coat shall be the manufacturer's standard color.

3.11.1 Cleaning. Metal surfaces to be painted shall be cleaned to insure that they are free from all oil, grease, welding slag and spatter, all mill scale, products of corrosion, dirt, or other foreign substances.

3.11.2 Treatment. As soon as practicable after cleaning, and before any evidence of rust or other contamination can result, metal surfaces shall be treated with a primer pretreatment coating to increase the adhesion of the coating system.

3.11.3 Painting. For metal surfaces, painting shall consist of at least one coat of rust-inhibiting primer and one coat of finish enamel. The rust-inhibiting primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes except that the total dry film thickness shall be not less than 2.5 mils. The paint shall be free from runs, sags, orange peel, or other defects.

3.12 Operating instructions. Complete operating and maintenance instructions, schematic diagrams, and parts list shall be provided with each public address set (see 6.2.2).

3.13 Accessories. When specified (see 6.2.1), accessories, such as installation wiring, extra microphones, speakers, horns, etc. shall be furnished.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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* 4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.6).

* 4.3 Quality conformance inspection. Each public address set shall be examined in accordance with 4.4 and tested in accordance with 4.5.

* 4.4 Examination. Each public address set shall be examined for compliance with the requirements specified in Section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Tests. Prior to submission to the Government for acceptance and delivery, each public address set shall be tested for satisfactory performance and compliance with Section 3 of this specification, and the safety standards of UL813. When specified (see 6.2.1), the procedure and methods used for accomplishing the tests shall be submitted in a written test plan by the manufacturers to the Government approval. Any public address set failing to pass any test in the accepted test plan shall be rejected. Certification of the manufacturer's test data, and the UL listing mark will be acceptable in lieu of tests (see 6.2.2).

4.6 Packaging inspection. The preservation, packaging, packing, and marking of the public address sets shall be inspected to verify conformance to the requirements of Section 5.

5. PACKAGING

5.1 Preservation-packaging. The preservation-packaging shall be level A or C, as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Amplifiers. The cabinet, with the amplifier and controls, shall be preserved in accordance with MIL-P-116, method IIb. Where practicable, the cabinet may be used as the inner container, provided sharp corners and projections are adequately cushioned against rupture of the barrier. Where it is not practicable to use the cabinet as the inner container, a fiberboard box conforming to PPP-B-636, class domestic, shall be used as the inner container, with the corners rounded or blunted to protect the barrier against rupture. The outer container shall be a fiberboard box conforming to pPP-B-636, W5c or W6c.

5.1.1.2 Loudspeakers. The loudspeakers shall be individually preserved in accordance with MIL-P-116, method IC-2.

5.1.1.3 Microphone. The microphone shall be individually preserved in accordance with MIL-P-116, method IA.

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5.1.1.4 Microphone stand. The mircrophone stand shall be individually preserved in accordance with MIL-P-226, method III.

5.1.1.5 Technical publications. Technical publications for each public address set shall be preserved in accordance with MIL-P-116, method IC-3.

5.1.1.6 Repair parts. Repair parts shall be preserved and packaged in accordance with MIL-E-17555 or in accordance with MIL-P-116 for parts not specifically covered in MIL-E-17555.

5.1.1.7 Intermediate packaging. Repair parts and publications shall be intermediate packaged in close-fitting boxes conforming to PPP-B-636, class domestic., The contents shall be cushioned and secured to prevent movement within the boxes.

5.1.2 Level C. The complete public address set shall be preserved and packaged in accordance with the supplier's standard practice.

5.2 Packing. The packing shall be level A, B, or C as specified (see 6.2.1).

5.2.1 Level A. Each complete public address set shall be packed in a close-fitting box conforming to PPP-B-601, overseas type. The contents shall be cushioned, blocked, and braced to prevent movement inside the container or damaging of contents.

5.2.2 Level B. Each complete public address set shall be packed as specified for level A, except that the boxes shall be domestic type.

5.2.3 Level C. The complete public address set shall be packed in a manner which will insure arrival at destination in satisfactory condition, and which will be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules, or with National Motor Freight Classification rules.

5.3 Marking. Interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The public address set is intended for installation in administrative and living quarter areas of advanced base encampments to be used for general announcements, indoor talk-back paging, and to entertain or address personnel indoors or out.

6.2 Ordering data. Procurement documents should specify:

* 6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.
(b) Power rating of amplifier, speakers, and drivers (see 3.1).
(c) When fungus resistance is not required (see 3.10).
(d) When color is other than specified (see 3.11).
(e) When installation wiring, extra speakers, horns, microphones are requested (see 3.13).

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(f) When test plan is required (see 4.5).
(g) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

6.3 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulation (DAR), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data specified below shall be delivered in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data requirements	Applicable DD 1664
3.12	Commercial Manuals	DI-M-2071
4.5	Certificate of Compliance	DI-E-2121

(Copies of Data Item Descriptions required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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